

Sealed Experiment Record

Sealed 2026-10-10 · Signed by the K-Veritas server
Session 55047e28 · Data hash 8e2b...eadc

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-10 20:48 and 20:52 UTC. Reported results:
fused_linear_cross_entropy_full_memory_baseline_over_liger 1.23818,
fused_linear_cross_entropy_full_speed_baseline_over_liger 0.275131,
rms_norm_full_memory_baseline_over_liger 4.70807, rms_norm_full_speed_baseline_over_liger 8.96247.
1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	55047e28-75c3-43a0-90e9-37a9ba957c40
Initialized	2026-10-10T20:48:44Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	1d8c165246d86e19501f2d4c5cf319398c31c0bb85a7d0173cb3ab081998aa46
Disclosure	open
Source code, run 1	578c8bbe770da3d39e5b2412218af574fcefef88d2e4a3cba02ba8ab3ddaff0af

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	3 minutes 21 seconds	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
fused_linear_cross_entropy_backward_spe...	5.34787		explicit
fused_linear_cross_entropy_forward_spee...	0.0909073		explicit
fused_linear_cross_entropy_full_memory_...	1.23818		explicit
fused_linear_cross_entropy_full_speed_ba...	0.275131		explicit
rms_norm_backward_speed_baseline_over...	15.0434		explicit
rms_norm_forward_speed_baseline_over_l...	7.91171		explicit
rms_norm_full_memory_baseline_over_li...	4.70807		explicit
rms_norm_full_speed_baseline_over_liger	8.96247		explicit
swiglu_backward_memory_baseline_over...	1.12276		explicit
swiglu_backward_speed_baseline_over_li...	1.01894		explicit
swiglu_forward_memory_baseline_over_li...	1.1035		explicit
swiglu_forward_speed_baseline_over_liger	1.01625		explicit
swiglu_full_memory_baseline_over_liger	1.12276		explicit
swiglu_full_speed_baseline_over_liger	1.01778		explicit
rows_measured	92		explicit

Inline claim	Value	Phase	Output line
fused_linear_cross_entropy_full_memory_...	1.23818		7
fused_linear_cross_entropy_full_speed_ba...	0.275131		9
rms_norm_full_memory_baseline_over_li...	4.70807		13
rms_norm_full_speed_baseline_over_liger	8.96247		15
swiglu_full_memory_baseline_over_liger	1.12276		21
swiglu_full_speed_baseline_over_liger	1.01778		23

4 Execution evidence

Coherence

Coherent (0.47): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	1986	145.8 s	2.68 GB	100%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 552 tracked files, 5 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	20:48:44	b235358c3a28695d	552
4	run_end	20:52:06	b235358c3a28695d	0

3 snapshots with no file change omitted here; full timeline in Appendix B.

Activity: 554 files read, 1 files written, 6 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-10T20:52:07.635128+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	8e2bf2961fc0da8e721b6f9c0eec4508e92617114d33f4893e7cd6d64efeeadc
Code bundle hash	6ab2beb6c789f698104452416e4907dda8c32e277b63a7fea484b3285a475c33

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
fused_linear_cross_entropy_backward_spe...	5.34787		4	explicit
fused_linear_cross_entropy_forward_spee...	0.0909073		5	explicit
fused_linear_cross_entropy_full_memory_...	1.23818		6	explicit
fused_linear_cross_entropy_full_speed_ba...	0.275131		8	explicit
rms_norm_backward_speed_baseline_over...	15.0434		10	explicit
rms_norm_forward_speed_baseline_over_l...	7.91171		11	explicit
rms_norm_full_memory_baseline_over_li...	4.70807		12	explicit
rms_norm_full_speed_baseline_over_liger	8.96247		14	explicit
swiglu_backward_memory_baseline_over...	1.12276		16	explicit
swiglu_backward_speed_baseline_over_li...	1.01894		17	explicit
swiglu_forward_memory_baseline_over_li...	1.1035		18	explicit
swiglu_forward_speed_baseline_over_liger	1.01625		19	explicit
swiglu_full_memory_baseline_over_liger	1.12276		20	explicit
swiglu_full_speed_baseline_over_liger	1.01778		22	explicit
rows_measured	92		24	explicit

Phase	Line	Time	CPU time	Memory	GPU
fused_linear_cross_entropy	1	20:48:44	1.5 s	0.62 GB	
rms_norm	2	20:51:09	0.0 s	0.02 GB	
swiglu	3	20:51:16	0.1 s	0.02 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	20:48:44	b235358c3a28695d	552
1	phase fused_linear_cross_entropy	20:48:44	b235358c3a28695d	0
2	phase rms_norm	20:51:09	b235358c3a28695d	0
3	phase swiglu	20:51:16	b235358c3a28695d	0
4	run_end	20:52:06	b235358c3a28695d	0

Event	Path or command	Hash
read	kv_run.sh	
read	benchmark/scripts/benchmark_fused_linear_cross_entropy.py	
read	.github/ISSUE_TEMPLATE/bug_report.yaml	
read	.github/ISSUE_TEMPLATE/feature_request.yaml	
read	.github/workflows/amd-ci.yml	
read	.github/workflows/benchmark.yml	
read	.github/workflows/checkstyle.yml	
read	.github/workflows/docs.yml	
read	.github/workflows/intel-ci.yml	
read	.github/workflows/merge-queue-failure.yml	
read	.github/workflows/nvi-ci.yml	

Event	Path or command	Hash
read	.github/workflows/publish-liger-cute-kernels.yml	
read	.github/workflows/publish-nightly.yml	
read	.github/workflows/publish-release.yml	
read	.pre-commit-config.yaml	
read	benchmark/__init__.py	
read	benchmark/benchmarks_visualizer.py	
read	benchmark/scripts/__init__.py	
read	benchmark/scripts/benchmark_attn_res.py	
read	benchmark/scripts/benchmark_cpo_loss.py	
read	benchmark/scripts/benchmark_cross_entropy.py	
read	benchmark/scripts/benchmark_cutedsl_fused_linear_cross_entropy_sm90.py	
read	benchmark/scripts/benchmark_cutile_fused_linear_cross_entropy_sm90.py	
read	benchmark/scripts/benchmark_distill_cosine_loss.py	
read	benchmark/scripts/benchmark_distill_jsd_loss.py	
read	benchmark/scripts/benchmark_dpo_loss.py	
read	benchmark/scripts/benchmark_dyt.py	
read	benchmark/scripts/benchmark_embedding.py	
read	benchmark/scripts/benchmark_fused_add_rms_norm.py	
read	benchmark/scripts/benchmark_fused_linear_jsd.py	
read	benchmark/scripts/benchmark_fused_linear_kl_div.py	
read	benchmark/scripts/benchmark_fused_moe.py	
read	benchmark/scripts/benchmark_fused_neighborhood_attention.py	
read	benchmark/scripts/benchmark_fused_scaled_cross_entropy_sm90.py	
read	benchmark/scripts/benchmark_fused_scaled_linear_cross_entropy_tp.py	
read	benchmark/scripts/benchmark_geglu.py	
read	benchmark/scripts/benchmark_group_norm.py	
read	benchmark/scripts/benchmark_grpo_loss.py	
read	benchmark/scripts/benchmark_grpo_loss_kernel.py	
read	benchmark/scripts/benchmark_jsd.py	
read	benchmark/scripts/benchmark_kl_div.py	
read	benchmark/scripts/benchmark_kto_loss.py	
read	benchmark/scripts/benchmark_layer_norm.py	
read	benchmark/scripts/benchmark_llama4_rope.py	
read	benchmark/scripts/benchmark_megatron_cross_entropy.py	
read	benchmark/scripts/benchmark_megatron_rms_norm.py	
read	benchmark/scripts/benchmark_megatron_swiglu.py	
read	benchmark/scripts/benchmark_mhc.py	
read	benchmark/scripts/benchmark_mhc_lm.py	
read	benchmark/scripts/benchmark_mlp.py	
read	benchmark/scripts/benchmark_modulated_rms_norm.py	
read	benchmark/scripts/benchmark_multi_token_attention.py	
read	benchmark/scripts/benchmark_orpo_loss.py	
read	benchmark/scripts/benchmark_poly_norm.py	
read	benchmark/scripts/benchmark_qwen2vl_mrope.py	
read	benchmark/scripts/benchmark_relu_squared.py	
read	benchmark/scripts/benchmark_rms_norm.py	
read	benchmark/scripts/benchmark_rope.py	
read	benchmark/scripts/benchmark_simpo_loss.py	
read	benchmark/scripts/benchmark_softmax.py	
read	benchmark/scripts/benchmark_sparse_multi_token_attention.py	

Event	Path or command	Hash
read	benchmark/scripts/benchmark_sparsemax.py	
read	benchmark/scripts/benchmark_swiglu.py	
read	benchmark/scripts/benchmark_tiled_mlp.py	
read	benchmark/scripts/benchmark_tvd.py	
read	benchmark/scripts/run_cutedsl_compare.py	
read	benchmark/scripts/run_cutile_compare.py	
read	benchmark/scripts/utils.py	
read	dev/modal/benchmarks.py	
read	dev/modal/tests.py	
read	examples/alignment/accelerate_config.yaml	
read	examples/alignment/run_orpo.py	
read	examples/huggingface/callback.py	
read	examples/huggingface/config/fsdp_config.json	
read	examples/huggingface/launch_on_modal.py	
read	examples/huggingface/run_benchmarks.sh	
read	examples/huggingface/run_gemma.sh	
read	examples/huggingface/run_llama.sh	
read	examples/huggingface/run_qwen.sh	
read	examples/huggingface/run_qwen2_vl.sh	
read	examples/huggingface/training.py	
read	examples/huggingface/training_multimodal.py	
read	examples/lightning/training.py	
read	examples/medusa/callback.py	
read	examples/medusa/medusa_util.py	
read	examples/medusa/scripts/llama3_8b_medusa.sh	
read	examples/medusa/train.py	
read	examples/megatron/run_mode1_monkey_patch.py	
read	examples/megatron/run_mode2_hand_spec.py	
read	liger_cute_kernels/cmake/export_nonrdc_nvshmem_ptx.py	
read	liger_cute_kernels/csrc/core/include/liger_cute/check.h	
read	liger_cute_kernels/csrc/core/include/liger_cute/detail/status.h	
read	liger_cute_kernels/csrc/core/include/liger_cute/detail/symmetric_memory.h	
read	liger_cute_kernels/csrc/core/include/liger_cute/export.h	
read	liger_cute_kernels/csrc/core/include/liger_cute/liger_cute.h	
read	liger_cute_kernels/csrc/core/include/liger_cute/moe.h	
read	liger_cute_kernels/csrc/core/include/liger_cute/nvshmem.h	
read	liger_cute_kernels/csrc/core/src/fused_scaled_linear_cross_entropy/state.h	
read	liger_cute_kernels/csrc/core/src/moe/moe_context.h	
read	liger_cute_kernels/csrc/core/src/moe/moe_fwd_bwd_tune_configs.hpp	
read	liger_cute_kernels/csrc/core/src/moe/moe_fwd_bwd_tune_configs_sm100.hpp	
read	liger_cute_kernels/csrc/core/src/moe/moe_fwd_bwd_tune_configs_sm90.hpp	
read	liger_cute_kernels/csrc/core/src/moe/moe_launch.h	
read	liger_cute_kernels/csrc/core/src/moe/moe_nonrdc_module.cpp	
read	liger_cute_kernels/csrc/core/src/moe/moe_nonrdc_module.h	
read	liger_cute_kernels/cute_build.py	
read	liger_cute_kernels/liger_cute_kernels/__init__.py	
read	liger_cute_kernels/liger_cute_kernels/configuration.py	
read	liger_cute_kernels/liger_cute_kernels/nvshmem.py	
read	liger_cute_kernels/liger_cute_kernels/tvm_ffi.py	
read	liger_cute_kernels/liger_cute_kernels/tvm_ffi_bindings.cpp	

Event	Path or command	Hash
read	liger_cute_kernels/pyproject.toml	
read	liger_cute_kernels/setup.py	
read	liger_cute_kernels/test/test_configuration.py	
read	liger_cute_kernels/test/test_cute_build.py	
read	liger_cute_kernels/test/test_export_nonrdc_nvshmem_ptx.py	
read	liger_cute_kernels/test/test_fslce_teams.py	
read	liger_cute_kernels/test/test_moe_architecture_split.py	
read	liger_cute_kernels/test/test_moe_bindings.py	
read	liger_cute_kernels/test/test_moe_cuda_graph.py	
read	liger_cute_kernels/test/test_moe_team_setup.py	
read	liger_cute_kernels/test/test_nvshmem.py	
read	liger_cute_kernels/test/test_tvm_ffi_loader.py	
read	liger_cute_kernels/tests/cpp/test_moe_api_dispatch.cpp	
read	liger_cute_kernels/tests/python/benchmark_fused_scaled_linear_cross_entrop...	
read	liger_cute_kernels/tests/python/benchmark_fused_scaled_linear_cross_entrop...	
read	liger_cute_kernels/tests/python/fslce_tp_test_utils.py	
read	liger_cute_kernels/tests/python/validate_fused_scaled_linear_cross_entropy_...	
read	mkdocs.yml	
read	pyproject.toml	
read	setup.py	
read	src/liger_kernel/backends/capability.py	
read	src/liger_kernel/backends/dispatch.py	
read	src/liger_kernel/backends/registry.py	
read	src/liger_kernel/chunked_loss/__init__.py	
read	src/liger_kernel/chunked_loss/cosine_similarity_loss.py	
read	src/liger_kernel/chunked_loss/cpo_loss.py	
read	src/liger_kernel/chunked_loss/dpo_loss.py	
read	src/liger_kernel/chunked_loss/functional.py	
read	src/liger_kernel/chunked_loss/fused_linear_distillation.py	
read	src/liger_kernel/chunked_loss/fused_linear_ppo.py	
read	src/liger_kernel/chunked_loss/fused_linear_preference.py	
read	src/liger_kernel/chunked_loss/fused_linear_unpaired_preference.py	
read	src/liger_kernel/chunked_loss/grpo_loss.py	
read	src/liger_kernel/chunked_loss/jsd_loss.py	
read	src/liger_kernel/chunked_loss/kto_loss.py	
read	src/liger_kernel/chunked_loss/orpo_loss.py	
read	src/liger_kernel/chunked_loss/simpo_loss.py	
read	src/liger_kernel/env_report.py	
read	src/liger_kernel/megatron/__init__.py	
read	src/liger_kernel/megatron/cross_entropy.py	
read	src/liger_kernel/megatron/monkey_patch.py	
read	src/liger_kernel/megatron/rms_norm.py	
read	src/liger_kernel/megatron/swiglu.py	
read	src/liger_kernel/ops/_nvidia_shared.py	
read	src/liger_kernel/ops/attn_res.py	
read	src/liger_kernel/ops/backends/__init__.py	
read	src/liger_kernel/ops/backends/_ascend/__init__.py	
read	src/liger_kernel/ops/backends/_ascend/cross_entropy.py	
read	src/liger_kernel/ops/backends/_ascend/dispatch_common.py	
read	src/liger_kernel/ops/backends/_ascend/fused_linear_cross_entropy.py	

Event	Path or command	Hash
read	src/liger_kernel/ops/backends/_ascend/layer_norm.py	
read	src/liger_kernel/ops/backends/_ascend/ops/__init__.py	
read	src/liger_kernel/ops/backends/_ascend/ops/attn_res.py	
read	src/liger_kernel/ops/backends/_ascend/ops/cross_entropy.py	
read	src/liger_kernel/ops/backends/_ascend/ops/dyt.py	
read	src/liger_kernel/ops/backends/_ascend/ops/embedding.py	
read	src/liger_kernel/ops/backends/_ascend/ops/fused_add_rms_norm.py	
read	src/liger_kernel/ops/backends/_ascend/ops/fused_linear_cross_entropy.py	
read	src/liger_kernel/ops/backends/_ascend/ops/fused_linear_jsd.py	
read	src/liger_kernel/ops/backends/_ascend/ops/fused_moe.py	
read	src/liger_kernel/ops/backends/_ascend/ops/fused_moe_kernels.py	
read	src/liger_kernel/ops/backends/_ascend/ops/fused_neighborhood_attention.py	
read	src/liger_kernel/ops/backends/_ascend/ops/geglu.py	
read	src/liger_kernel/ops/backends/_ascend/ops/group_norm.py	
read	src/liger_kernel/ops/backends/_ascend/ops/grpo_loss.py	
read	src/liger_kernel/ops/backends/_ascend/ops/jsd.py	
read	src/liger_kernel/ops/backends/_ascend/ops/kl_div.py	
read	src/liger_kernel/ops/backends/_ascend/ops/layer_norm.py	
read	src/liger_kernel/ops/backends/_ascend/ops/llama4_rope.py	
read	src/liger_kernel/ops/backends/_ascend/ops/mhc.py	
read	src/liger_kernel/ops/backends/_ascend/ops/poly_norm.py	
read	src/liger_kernel/ops/backends/_ascend/ops/qwen2vl_mrope.py	
read	src/liger_kernel/ops/backends/_ascend/ops/rms_norm.py	
read	src/liger_kernel/ops/backends/_ascend/ops/rope.py	
read	src/liger_kernel/ops/backends/_ascend/ops/softmax.py	
read	src/liger_kernel/ops/backends/_ascend/ops/sparsemax.py	
read	src/liger_kernel/ops/backends/_ascend/ops/swiglu.py	
read	src/liger_kernel/ops/backends/_ascend/ops/tvd.py	
read	src/liger_kernel/ops/backends/_ascend/rms_norm.py	
read	src/liger_kernel/ops/backends/_ascend/rope.py	
read	src/liger_kernel/ops/backends/_ascend/swiglu.py	
read	src/liger_kernel/ops/backends/_ascend/ub_manager.py	
read	src/liger_kernel/ops/backends/_cutedsl/_cute_lib/__init__.py	
read	src/liger_kernel/ops/backends/_cutedsl/_cute_lib/compile_utils.py	
read	src/liger_kernel/ops/backends/_cutedsl/_cute_lib/copy_utils.py	
read	src/liger_kernel/ops/backends/_cutedsl/_cute_lib/dtype_map.py	
read	src/liger_kernel/ops/backends/_cutedsl/_cute_lib/layout_utils.py	
read	src/liger_kernel/ops/backends/_cutedsl/_cute_lib/reduce.py	
read	src/liger_kernel/ops/backends/_cutedsl/_cute_lib/reduction_base.py	
read	src/liger_kernel/ops/backends/_cutedsl/_cute_lib/rmsnorm_fwd.py	
read	src/liger_kernel/ops/backends/_cutedsl/_cute_lib/utils.py	
read	src/liger_kernel/ops/backends/_cutedsl/cross_entropy.py	
read	src/liger_kernel/ops/backends/_cutedsl/fused_add_rms_norm.py	
read	src/liger_kernel/ops/backends/_cutedsl/fused_linear_cross_entropy.py	
read	src/liger_kernel/ops/backends/_cutedsl/fused_linear_jsd.py	
read	src/liger_kernel/ops/backends/_cutedsl/geglu.py	
read	src/liger_kernel/ops/backends/_cutedsl/jsd.py	
read	src/liger_kernel/ops/backends/_cutedsl/kl_div.py	
read	src/liger_kernel/ops/backends/_cutedsl/layer_norm.py	
read	src/liger_kernel/ops/backends/_cutedsl/rms_norm.py	

Event	Path or command	Hash
read	src/liger_kernel/ops/backends/_cuteds/rope.py	
read	src/liger_kernel/ops/backends/_cuteds/softmax.py	
read	src/liger_kernel/ops/backends/_cuteds/swiglu.py	
read	src/liger_kernel/ops/backends/_cutile/__init__.py	
read	src/liger_kernel/ops/backends/_cutile/fused_linear_jsd.py	
read	src/liger_kernel/ops/backends/_cutile/jsd.py	
read	src/liger_kernel/ops/backends/_cutile/layer_norm.py	
read	src/liger_kernel/ops/backends/_cutile/rms_norm.py	
read	src/liger_kernel/ops/backends/_cutile/softmax.py	
read	src/liger_kernel/ops/backends/_triton/__init__.py	
read	src/liger_kernel/ops/backends/_triton/cross_entropy.py	
read	src/liger_kernel/ops/backends/_triton/fused_add_rms_norm.py	
read	src/liger_kernel/ops/backends/_triton/fused_linear_cross_entropy.py	
read	src/liger_kernel/ops/backends/_triton/fused_linear_jsd.py	
read	src/liger_kernel/ops/backends/_triton/geglu.py	
read	src/liger_kernel/ops/backends/_triton/jsd.py	
read	src/liger_kernel/ops/backends/_triton/kl_div.py	
read	src/liger_kernel/ops/backends/_triton/layer_norm.py	
read	src/liger_kernel/ops/backends/_triton/rms_norm.py	
read	src/liger_kernel/ops/backends/_triton/rope.py	
read	src/liger_kernel/ops/backends/_triton/softmax.py	
read	src/liger_kernel/ops/backends/_triton/swiglu.py	
read	src/liger_kernel/ops/backends/registry.py	
read	src/liger_kernel/ops/configure.py	
read	src/liger_kernel/ops/cross_entropy.py	
read	src/liger_kernel/ops/cute/__init__.py	
read	src/liger_kernel/ops/cute/fused_linear_scaled_cross_entropy_tp.py	
read	src/liger_kernel/ops/cute/ops/__init__.py	
read	src/liger_kernel/ops/cute/ops/moe.py	
read	src/liger_kernel/ops/cuteds/ops/__init__.py	
read	src/liger_kernel/ops/cuteds/ops/_fused_linear_cross_entropy_backward_sm9...	
read	src/liger_kernel/ops/cuteds/ops/_fused_linear_cross_entropy_forward_sm90....	
read	src/liger_kernel/ops/cuteds/ops/_fused_scaled_cross_entropy_backward_fuse...	
read	src/liger_kernel/ops/cuteds/ops/_fused_scaled_cross_entropy_forward_fragm...	
read	src/liger_kernel/ops/cuteds/ops/_fused_scaled_cross_entropy_utils_sm90.py	
read	src/liger_kernel/ops/cuteds/ops/_sm100_gemm.py	
read	src/liger_kernel/ops/cuteds/ops/_sm90_persistent_gemm.py	
read	src/liger_kernel/ops/cuteds/ops/cross_entropy.py	
read	src/liger_kernel/ops/cuteds/ops/fused_linear_cross_entropy.py	
read	src/liger_kernel/ops/cuteds/ops/fused_linear_cross_entropy_sm90.py	
read	src/liger_kernel/ops/cuteds/ops/fused_scaled_cross_entropy_sm90.py	
read	src/liger_kernel/ops/cuteds/ops/rms_norm.py	
read	src/liger_kernel/ops/cuteds/ops/rms_norm_dw_reduce.py	
read	src/liger_kernel/ops/cuteds/ops/rms_norm_fastpath.py	
read	src/liger_kernel/ops/cuteds/ops/rope.py	
read	src/liger_kernel/ops/cuteds/ops/swiglu.py	
read	src/liger_kernel/ops/cuteds/ops/utils.py	
read	src/liger_kernel/ops/cutile/ops/__init__.py	
read	src/liger_kernel/ops/cutile/ops/cross_entropy.py	
read	src/liger_kernel/ops/cutile/ops/dyt.py	

Event	Path or command	Hash
read	src/liger_kernel/ops/cutile/ops/fused_add_rms_norm.py	
read	src/liger_kernel/ops/cutile/ops/fused_linear_cross_entropy.py	
read	src/liger_kernel/ops/cutile/ops/fused_linear_jsd.py	
read	src/liger_kernel/ops/cutile/ops/fused_linear_scaled_cross_entropy.py	
read	src/liger_kernel/ops/cutile/ops/geglu.py	
read	src/liger_kernel/ops/cutile/ops/group_norm.py	
read	src/liger_kernel/ops/cutile/ops/grpo_loss.py	
read	src/liger_kernel/ops/cutile/ops/jsd.py	
read	src/liger_kernel/ops/cutile/ops/kl_div.py	
read	src/liger_kernel/ops/cutile/ops/layer_norm.py	
read	src/liger_kernel/ops/cutile/ops/llama4_rope.py	
read	src/liger_kernel/ops/cutile/ops/multi_token_attention.py	
read	src/liger_kernel/ops/cutile/ops/poly_norm.py	
read	src/liger_kernel/ops/cutile/ops/qwen2vl_mrope.py	
read	src/liger_kernel/ops/cutile/ops/rms_norm.py	
read	src/liger_kernel/ops/cutile/ops/rope.py	
read	src/liger_kernel/ops/cutile/ops/softmax.py	
read	src/liger_kernel/ops/cutile/ops/sparsemax.py	
read	src/liger_kernel/ops/cutile/ops/swiglu.py	
read	src/liger_kernel/ops/cutile/ops/tiled_mlp.py	
read	src/liger_kernel/ops/cutile/ops/utils.py	
read	src/liger_kernel/ops/experimental/embedding.py	
read	src/liger_kernel/ops/experimental/mm_int8int2.py	
read	src/liger_kernel/ops/fused_add_rms_norm.py	
read	src/liger_kernel/ops/fused_ce_tvd.py	
read	src/liger_kernel/ops/fused_linear_scaled_cross_entropy.py	
read	src/liger_kernel/ops/group_norm.py	
read	src/liger_kernel/ops/grpo_loss.py	
read	src/liger_kernel/ops/llama4_rope.py	
read	src/liger_kernel/ops/poly_norm.py	
read	src/liger_kernel/ops/relu_squared.py	
read	src/liger_kernel/ops/rope.py	
read	src/liger_kernel/ops/sparsemax.py	
read	src/liger_kernel/ops/swiglu.py	
read	src/liger_kernel/ops/utils.py	
read	src/liger_kernel/testing/__init__.py	
read	src/liger_kernel/testing/_op_helpers.py	
read	src/liger_kernel/transformers/auto_model.py	
read	src/liger_kernel/transformers/dyt.py	
read	src/liger_kernel/transformers/experimental/__init__.py	
read	src/liger_kernel/transformers/experimental/embedding.py	
read	src/liger_kernel/transformers/fsdp.py	
read	src/liger_kernel/transformers/fused_add_rms_norm.py	
read	src/liger_kernel/transformers/fused_ce_tvd.py	
read	src/liger_kernel/transformers/fused_linear_cross_entropy.py	
read	src/liger_kernel/transformers/fused_linear_jsd.py	
read	src/liger_kernel/transformers/fused_linear_kl_div.py	
read	src/liger_kernel/transformers/fused_neighborhood_attention.py	
read	src/liger_kernel/transformers/group_norm.py	
read	src/liger_kernel/transformers/grpo_loss.py	

Event	Path or command	Hash
read	src/liger_kernel/transformers/jsd.py	
read	src/liger_kernel/transformers/kl_div.py	
read	src/liger_kernel/transformers/model/__init__.py	
read	src/liger_kernel/transformers/model/deepseek_v2.py	
read	src/liger_kernel/transformers/model/deepseek_v3.py	
read	src/liger_kernel/transformers/model/deepseek_v4.py	
read	src/liger_kernel/transformers/model/evaone4.py	
read	src/liger_kernel/transformers/model/falcon_h1.py	
read	src/liger_kernel/transformers/model/gemma.py	
read	src/liger_kernel/transformers/model/gemma2.py	
read	src/liger_kernel/transformers/model/gemma3.py	
read	src/liger_kernel/transformers/model/gemma4.py	
read	src/liger_kernel/transformers/model/glm4.py	
read	src/liger_kernel/transformers/model/glm4v.py	
read	src/liger_kernel/transformers/model/glm4v_moe.py	
read	src/liger_kernel/transformers/model/gpt_oss.py	
read	src/liger_kernel/transformers/model/hunyuan_v1.py	
read	src/liger_kernel/transformers/model/internvl.py	
read	src/liger_kernel/transformers/model/llama.py	
read	src/liger_kernel/transformers/model/llama4.py	
read	src/liger_kernel/transformers/model/llava.py	
read	src/liger_kernel/transformers/model/loss_utils.py	
read	src/liger_kernel/transformers/model/ministral.py	
read	src/liger_kernel/transformers/model/mistral.py	
read	src/liger_kernel/transformers/model/mixtral.py	
read	src/liger_kernel/transformers/model/mlama.py	
read	src/liger_kernel/transformers/model/muse_glimmer.py	
read	src/liger_kernel/transformers/model/nemotron.py	
read	src/liger_kernel/transformers/model/olmo2.py	
read	src/liger_kernel/transformers/model/olmo3.py	
read	src/liger_kernel/transformers/model/output_classes.py	
read	src/liger_kernel/transformers/model/paligemma.py	
read	src/liger_kernel/transformers/model/phi3.py	
read	src/liger_kernel/transformers/model/pixtral.py	
read	src/liger_kernel/transformers/model/qwen2.py	
read	src/liger_kernel/transformers/model/qwen2_5_vl.py	
read	src/liger_kernel/transformers/model/qwen2_vl.py	
read	src/liger_kernel/transformers/model/qwen3.py	
read	src/liger_kernel/transformers/model/qwen3_5.py	
read	src/liger_kernel/transformers/model/qwen3_5_moe.py	
read	src/liger_kernel/transformers/model/qwen3_moe.py	
read	src/liger_kernel/transformers/model/qwen3_next.py	
read	src/liger_kernel/transformers/model/qwen3_vl.py	
read	src/liger_kernel/transformers/model/qwen3_vl_moe.py	
read	src/liger_kernel/transformers/model/smollm3.py	
read	src/liger_kernel/transformers/model/molvlm.py	
read	src/liger_kernel/transformers/modulated_rms_norm.py	
read	src/liger_kernel/transformers/moe.py	
read	src/liger_kernel/transformers/monkey_patch.py	
read	src/liger_kernel/transformers/multi_token_attention.py	

Event	Path or command	Hash
read	src/liger_kernel/transformers/poly_norm.py	
read	src/liger_kernel/transformers/qwen2vl_mrope.py	
read	src/liger_kernel/transformers/relu_squared.py	
read	src/liger_kernel/transformers/rms_norm.py	
read	src/liger_kernel/transformers/softmax.py	
read	src/liger_kernel/transformers/swiglu.py	
read	src/liger_kernel/transformers/trainer/__init__.py	
read	src/liger_kernel/transformers/trainer/orpo_trainer.py	
read	src/liger_kernel/transformers/trainer_integration.py	
read	src/liger_kernel/transformers/tvd.py	
read	src/liger_kernel/transformers/vocab_parallel_cross_entropy.py	
read	src/liger_kernel/triton/__init__.py	
read	src/liger_kernel/triton/monkey_patch.py	
read	test/__init__.py	
read	test/backends/__init__.py	
read	test/backends/test_capability.py	
read	test/backends/test_capability_device.py	
read	test/backends/test_cutile_select_mode.py	
read	test/backends/test_registry.py	
read	test/chunked_loss/__init__.py	
read	test/chunked_loss/test_cosine_loss.py	
read	test/chunked_loss/test_cpo_loss.py	
read	test/chunked_loss/test_dpo_loss.py	
read	test/chunked_loss/test_grpo_loss.py	
read	test/chunked_loss/test_jsd_loss.py	
read	test/chunked_loss/test_kto_loss.py	
read	test/chunked_loss/test_orpo_loss.py	
read	test/chunked_loss/test_simpo_loss.py	
read	test/confstest.py	
read	test/convergence/__init__.py	
read	test/convergence/bf16/__init__.py	
read	test/convergence/bf16/test_mini_models.py	
read	test/convergence/bf16/test_mini_models_multimodal.py	
read	test/convergence/bf16/test_mini_models_with_logits.py	
read	test/convergence/fp32/__init__.py	
read	test/convergence/fp32/test_mini_models.py	
read	test/convergence/fp32/test_mini_models_multimodal.py	
read	test/convergence/fp32/test_mini_models_with_logits.py	
read	test/cute/__init__.py	
read	test/cute/test_fused_linear_scaled_cross_entropy_tp.py	
read	test/cute/test_moe.py	
read	test/cute/test_moe_autograd.py	
read	test/cutedsl/__init__.py	
read	test/cutedsl/test_backend.py	
read	test/cutedsl/test_cross_entropy.py	
read	test/cutedsl/test_fused_add_rms_norm.py	
read	test/cutedsl/test_fused_linear_cross_entropy.py	
read	test/cutedsl/test_fused_linear_cross_entropy_sm90.py	
read	test/cutedsl/test_fused_scaled_cross_entropy_forward_fragment_sm90.py	
read	test/cutedsl/test_fused_scaled_cross_entropy_sm90.py	

Event	Path or command	Hash
read	test/cutedsl/test_gap_ops.py	
read	test/cutedsl/test_rms_norm.py	
read	test/cutedsl/test_rms_norm_fastpath.py	
read	test/cutedsl/test_rms_norm_stream.py	
read	test/cutedsl/test_rope.py	
read	test/cutedsl/test_shape_fallbacks.py	
read	test/cutedsl/test_sm100_gemm.py	
read	test/cutedsl/test_swiglu.py	
read	test/cutile/__init__.py	
read	test/cutile/test_backend.py	
read	test/cutile/test_cutile_backends_parity.py	
read	test/cutile/test_fused_linear_cross_entropy_sm90.py	
read	test/cutile/test_fused_linear_scaled_cross_entropy.py	
read	test/cutile/test_rms_norm.py	
read	test/cutile/test_rope.py	
read	test/cutile/test_swiglu.py	
read	test/megatron/__init__.py	
read	test/megatron/test_cross_entropy.py	
read	test/megatron/test_monkey_patch.py	
read	test/megatron/test_rms_norm.py	
read	test/megatron/test_swiglu.py	
read	test/ops/__init__.py	
read	test/ops/conftest.py	
read	test/ops/test_cutedsl_shape_fallbacks.py	
read	test/ops/test_fused_add_rms_norm.py	
read	test/ops/test_fused_linear_cross_entropy.py	
read	test/ops/test_fused_linear_jsd.py	
read	test/ops/test_fused_linear_kl_div.py	
read	test/ops/test_geglu.py	
read	test/ops/test_jsd.py	
read	test/ops/test_jsd_chunk_ignore_regression.py	
read	test/ops/test_kl_div.py	
read	test/ops/test_layer_norm.py	
read	test/ops/test_norm_lora_parity.py	
read	test/ops/test_rms_norm.py	
read	test/ops/test_rope.py	
read	test/ops/test_softmax.py	
read	test/ops/test_swiglu.py	
read	test/resources/fake_configs/Google/Gemma3/gemma-3-4b-it/tokenizer_config...	
read	test/resources/fake_configs/Google/Gemma4/gemma-4-e4b-it/tokenizer_conf...	
read	test/resources/fake_configs/Google/Paligemma/paligemma-3b-pt-224/tokeniz...	
read	test/resources/fake_configs/HuggingFaceTB/SmolVLM2-256M-Video-Instru...	
read	test/resources/fake_configs/Llava/llava-1.5-7b-hf/preprocessor_config.json	
read	test/resources/fake_configs/Llava/llava-1.5-7b-hf/processor_config.json	
read	test/resources/fake_configs/Llava/llava-1.5-7b-hf/tokenizer_config.json	
read	test/resources/fake_configs/OpenGVLab/InternVL3-1B-hf/tokenizer_config.j...	
read	test/resources/fake_configs/Qwen/Qwen2-VL-7B-Instruct/tokenizer_config.js...	
read	test/resources/fake_configs/Qwen/Qwen2.5-VL-7B-Instruct/tokenizer_config....	
read	test/resources/fake_configs/Qwen/Qwen3-VL-4B-Instruct/tokenizer_config.js...	
read	test/resources/fake_configs/meta-llama/Llama-3.2-11B-Vision-Instruct/tokeni...	

Event	Path or command	Hash
read	test/resources/fake_configs/meta-llama/Llama-4-Scout-17B-16E-Instruct/toke...	
read	test/resources/fake_configs/meta-models/Muse-Glimmer-30B/tokenizer_confi...	
read	test/resources/scripts/generate_tokenized_dataset.py	
read	test/resources/tiny_shakespeare_tokenized/dataset_info.json	
read	test/resources/tiny_shakespeare_tokenized/state.json	
read	test/transformers/test_attn_res.py	
read	test/transformers/test_auto_model.py	
read	test/transformers/test_cross_entropy.py	
read	test/transformers/test_dyt.py	
read	test/transformers/test_embedding.py	
read	test/transformers/test_flex_attention.py	
read	test/transformers/test_fused_add_rms_norm.py	
read	test/transformers/test_fused_ce_tvd.py	
read	test/transformers/test_fused_linear_cross_entropy.py	
read	test/transformers/test_fused_linear_jsd.py	
read	test/transformers/test_fused_linear_kl_div.py	
read	test/transformers/test_fused_linear_scaled_cross_entropy.py	
read	test/transformers/test_fused_moe.py	
read	test/transformers/test_fused_neighborhood_attention.py	
read	test/transformers/test_geglu.py	
read	test/transformers/test_group_norm.py	
read	test/transformers/test_grpo_loss.py	
read	test/transformers/test_jsd.py	
read	test/transformers/test_kl_div.py	
read	test/transformers/test_layer_norm.py	
read	test/transformers/test_llama4.py	
read	test/transformers/test_llama4_rope.py	
read	test/transformers/test_mhc.py	
read	test/transformers/test_mlp.py	
read	test/transformers/test_mm_int8int2.py	
read	test/transformers/test_modulated_rms_norm.py	
read	test/transformers/test_monkey_patch.py	
read	test/transformers/test_multi_token_attention.py	
read	test/transformers/test_muse_glimmer.py	
read	test/transformers/test_poly_norm.py	
read	test/transformers/test_qwen2vl_mrope.py	
read	test/transformers/test_relu_squared.py	
read	test/transformers/test_rms_norm.py	
read	test/transformers/test_rope.py	
read	test/transformers/test_selective_log_softmax_strides.py	
read	test/transformers/test_softmax.py	
read	test/transformers/test_sparsemax.py	
read	test/transformers/test_swiglu.py	
read	test/transformers/test_tiled_mlp.py	
read	test/transformers/test_trainer_integration.py	
read	test/transformers/test_transformers.py	
read	test/transformers/test_tvd.py	
read	test/transformers/test_utils.py	
read	test/triton/test_triton_monkey_patch.py	
read	test/utils.py	

Event	Path or command	Hash
read	benchmark/scripts/benchmark_model_configs.py	80a0f0e9a3b0dceb
read	src/liger_kernel/__init__.py	
read	src/liger_kernel/functional.py	
read	src/liger_kernel/backends/__init__.py	
read	src/liger_kernel/utils.py	
read	src/liger_kernel/transformers/__init__.py	
read	src/liger_kernel/transformers/cross_entropy.py	
read	src/liger_kernel/ops/__init__.py	
read	src/liger_kernel.egg-info/top_level.txt	
read	src/liger_kernel.egg-info/PKG-INFO	
read	src/liger_kernel/ops/dyt.py	
read	src/liger_kernel/ops/fused_linear_cross_entropy.py	
read	src/liger_kernel/ops/fused_linear_jsd.py	
read	src/liger_kernel/ops/fused_linear_kl_div.py	
read	src/liger_kernel/ops/fused_moe.py	
read	src/liger_kernel/ops/fused_moe_kernels.py	
read	src/liger_kernel/ops/fused_neighborhood_attention.py	
read	src/liger_kernel/ops/softmax.py	
read	src/liger_kernel/ops/geglu.py	
read	src/liger_kernel/ops/jsd.py	
read	src/liger_kernel/ops/kl_div.py	
read	src/liger_kernel/ops/layer_norm.py	
read	src/liger_kernel/ops/mhc.py	
read	src/liger_kernel/ops/mlp.py	
read	src/liger_kernel/ops/modulated_rms_norm.py	
read	src/liger_kernel/ops/multi_token_attention.py	
read	src/liger_kernel/ops/qwen2vl_mrope.py	
read	src/liger_kernel/ops/rms_norm.py	
read	src/liger_kernel/ops/tiled_mlp.py	
read	src/liger_kernel/ops/tvd.py	
read	src/liger_kernel/ops/vocab_parallel_cross_entropy.py	
read	src/liger_kernel/ops/backends/_cutedsl/__init__.py	
read	src/liger_kernel/ops/cutedsl/__init__.py	
read	src/liger_kernel/ops/cutile/__init__.py	
read	src/liger_kernel/transformers/functional.py	
read	src/liger_kernel/transformers/geglu.py	
read	src/liger_kernel/transformers/layer_norm.py	
read	src/liger_kernel/transformers/llama4_rope.py	
read	src/liger_kernel/transformers/mhc.py	
read	src/liger_kernel/transformers/mlp.py	
read	src/liger_kernel/transformers/rope.py	
read	src/liger_kernel/transformers/sparsemax.py	
read	src/liger_kernel/transformers/tiled_mlp.py	
write	benchmark/data/all_benchmark_data.csv	
process	python benchmark/scripts/benchmark_fused_linear_cross_entropy.py	
process	pid 2907980	
process	python benchmark/scripts/benchmark_rms_norm.py	
process	file -b /var/tmp/kv-repro/035-linkedln-Liger-Kernel/env/bin/python3.12	
process	python benchmark/scripts/benchmark_swiglu.py	
process	python -	

C Run history

Invocation 1, OK, 3 minutes 21 seconds, 24 output lines, started 2026-10-10T20:48:44

```
metric_hash 748611fe985d8de3f2195aebded72b5ba6a50bedaa48c57774bef8b4689bcd24
run_digest  2f8d7788642c132ffa810c594512244be3c1bc148dfbb363d26046dee42f8d69
anchored    at run end
```

D Raw seal fields and hashes

Seal

```
data_hash      8e2bf2961fc0da8e721b6f9c0eec4508e92617114d33f4893e7cd6d64efeeadc
signed_message_hash bf96baa00a746ee7764ba8ef1e0ad90ac20627f942ad294634bb6518ff813ec1
nonce          f9bb6719985b7c5d029129e53f460964
signed_at      2026-10-10T20:52:07.635128+00:00
sealed_at      2026-10-10T20:52:07.694285902Z
checkout_bundle 6ab2beb6c789f698104452416e4907dda8c32e277b63a7fea484b3285a475c33
```

Signature (base64)

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MTgXJS1SnnYRqGxr0DLDRiWkbaRbXhJ6xHNwksrvmT2UyvEwCPVC/4RpMGGeCalagS1KG6o/K+EuvjUp8wlhz2n
qfy/Y/Xy1uHGM1qkyogjIITYoE6AHYV94y62BB6bovBcGxin4P5rF85izCFVPaPiqrVTqenoHFGpFk9jqTjBiJwS
AjYPkRWHNg4vQwaDT4q/flVYfUKf05rgft4RiY3tFTvQLwg8LT9Mpo1KCEZ0Qn2ebKLoL6p6igN4TgISyiSGXErn
J7H2qvcUB5H4VL0uSWwVUBGDWQB2WJ0RRDCuTYyp+bRX+9fSbrILp2b6Pq67Zn3F+I22l0B+QkLDD+nft/rxUo0D
IQKslzcyT0Uto46KucqteVxPU047T6Nin/so51ZbDhwoYrS4sNn5BktRk9KefYi50ezgg6j9JvsxSJYWIvrdL3m
WMHzB20ElhAlj2Wd5EEQ3PzX/BkBJz0tUMY19NmajhYcmYw9Prwm8tBLDWQiJULryrMY+XJMrUGZVgr/u66dLY9j
HnJyh3IPADBaivtkfxpDRU39oZU3luvQZr0Cz2CB0MZFaM5Zp7WyeN+pyXwTbTVz2dcT+YgA4azHTKNXFu0NVVWSx
TPJVDnuUxbUKLYWBhChE9F+PjMgJcbH4ZAiR6RK4D1VctoLUypKCYNm8KfzVhVzD78w=
```

Public key

```
-----BEGIN PUBLIC KEY-----
MIICijANBgkqhkiG9w0BAQEFAAACAg8AMIICCgKCAgEA1QL3YyMgeShx8IhTC7jb
sM0w2BnRXgsBn3+SMXKjtWATqdekmlPlcjayJxD5fcPrOPPCqp/LouFz5CUZzRi
6+8FTv70dHuIz5A4LAt4vcgwP+iWscPc3HzpmjpsKbTmOemqgp6RB6GNNppxYa5X
7qU95EfEisX/GfoS9ToxtgPBwCcVWSPBX8efec4TC58m96Fsp4mC8KicEadbIbHN
av0JjdTJ//CM0WZSIbdzE6qvEbNFCqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdHq+D
AhfTRUJNgWEkjdW5lb0Vhi4qlkweSOEMTKT3ye9Il5xEaJn4bku4eKqes7VSzt3o
XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
HNBPgPZ242NH8JgE6CNF7ToL9syasS67u7UMGIcNgeNHNcy9bLgpU5annzP9gN2l
6HEMldkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7lE3mbQRbPsLL/JApDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/oLCHBCf1VIDCkDxoehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v
8Jap/1PWjNcNku5cdw3gHLcCAwEAAQ==
-----END PUBLIC KEY-----
```

Run 1 outputs

```
stdout e0928ef5d07836bd4d21f5879d93260fdbb87ed5088bf8c3892b202784560564
stderr e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics 748611fe985d8de3f2195aebded72b5ba6a50bedaa48c57774bef8b4689bcd24
digest 2f8d7788642c132ffa810c594512244be3c1bc148dfbb363d26046dee42f8d69
```

kv_run.sh (unchanged)

```
pre 3bbcf796daf589836bcdad2a49967762368770e74315db1ae5c02f3ab976d7b3
post 3bbcf796daf589836bcdad2a49967762368770e74315db1ae5c02f3ab976d7b3
```

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,  
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},  
  author={Mamadou K. Keita and Christopher Homan},  
  year={2026},  
  eprint={2605.08586},  
  archivePrefix={arXiv},  
  primaryClass={cs.CR},  
  url={https://arxiv.org/abs/2605.08586},  
}
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